Lecture 37

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1 Infinite Limits

1.1 Divergence to Infinity

1.1.1 Definition

We say that a sequence (a_n) **diverges to** $+\infty$ if for every $M \in \mathbb{N}$, there exists an $N \in \mathbb{N}$ such that for all $n \geq N$:

$$a_n > M$$

[cite$_{\text{start}}$][cite: 440, 442, 443]

[cite$_{\text{start}}$]Similarly, (a_n) **diverges to** $-\infty$ if for every M, eventually $a_n < -M$[cite: 444, 446].

1.1.2 Notation

If (a_n) diverges to $+\infty$, we write:

$$\lim_{n\to\infty} a_n = +\infty$$

[cite$_{\text{start}}$][cite: 447, 449]

1.1.3 Distinction in Terminology

The term "divergent" has two meanings:

1. [cite$_{\text{start}}$]**General Divergence:** The sequence simply does not converge (e.g., $a_n = (-1)^n$ oscillates)[cite: 455].

2. [cite$_{\text{start}}$]**Proper Divergence:** The sequence goes to $+\infty$ or $-\infty$ (e.g., $a_n = n$)[cite: 456, 459].

1.2 The Extended Real Line

[cite$_{\text{start}}$]We define the extended real line as $\overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty, -\infty\}$[cite: 462].

1.2.1 Arithmetic Rules involving Infinity

1. **Addition:**

   - [cite$_{\text{start}}$]$a + (+\infty) = +\infty$ for any $a \in \mathbb{R}$[cite: 496].
   - [cite$_{\text{start}}$]$(+\infty) + (+\infty) = +\infty$[cite: 494].

- [cite$_{\text{start}}$]$(-\infty) + (+\infty)$ is **undefined**[cite: 494].

2. **Multiplication:**

$$a \cdot (+\infty) = \begin{cases} +\infty & \text{if } a > 0 \\ -\infty & \text{if } a < 0 \\ 0 & \text{if } a > 0 \text{ (context dependent, usually } 0 \cdot \infty \text{ is undefined)} \\ \text{undefined} & \text{if } a = 0 \end{cases}$$

[cite$_{\text{start}}$][cite: 495]

1.3 Relationship between Null Sequences and Infinity

Theorem: Let (a_n) be a positive sequence ($a_n > 0$). Then:

$$a_n \to 0 \iff \frac{1}{a_n} \to +\infty$$

[cite$_{\text{start}}$][cite: 464, 468, 470]

Proof Sketch: [cite$_{\text{start}}$]If a_n becomes arbitrarily small (positive), its reciprocal becomes arbitrarily large ($> M$)[cite: 460, 482].

1.4 Indeterminate Forms

One cannot simply subtract limits that are infinite. **Example:**

$$\lim_{n\to\infty} (\sqrt{n+1} - \sqrt{n})$$

- [cite$_{\text{start}}$]Naive attempt: $\lim \sqrt{n+1} - \lim \sqrt{n} = +\infty - (+\infty)$ which is undefined[cite: 478, 480].

- Correct approach using conjugation:

$$\sqrt{n+1} - \sqrt{n} = \frac{(\sqrt{n+1} - \sqrt{n})(\sqrt{n+1} + \sqrt{n})}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{\sqrt{n+1} + \sqrt{n}}$$

  [cite$_{\text{start}}$]As $n \to \infty$, the denominator $\to \infty$, so the limit is 0[cite: 487, 489].

2 Monotone Convergence Theorem (MCT)

2.1 Statement

1. [cite$_{\text{start}}$]Every **increasing** sequence that is **bounded above** converges[cite: 493, 497, 498].

2. [cite$_{\text{start}}$]Every **decreasing** sequence that is **bounded below** converges[cite: 499].

2.2 Proof (for Increasing Sequences)

Let $A = \{a_1, a_2, a_3, \dots\}$ be the set of terms. [cite$_{\text{start}}$]Let $a = \sup A$ (the least upper bound)[cite: 503, 504]. Since a is the supremum:

1. [cite$_{\text{start}}$]$a_n \leq a$ for all n[cite: 504].

2. [cite$_{\text{start}}$]For any $\epsilon > 0$, $a - \epsilon$ is not an upper bound, so there exists some N such that $a - \epsilon < a_N$[cite: 513].

Because (a_n) is increasing, for all $n \geq N$:

$$a - \epsilon < a_N \leq a_n \leq a < a + \epsilon$$

Thus, $|a_n - a|[cite_start] < \epsilon$, proving convergence[cite: 520, 523, 525].

3 Example: Recursive Sequence

Consider the sequence defined by:

$$x_1 = 2, \quad x_{n+1} = \frac{1}{2}\left(x_n + \frac{1}{x_n}\right)$$

[cite$_{\text{start}}$][cite: 533]

3.1 1. Calculate First Terms

- $x_1 = 2$
- [cite$_{\text{start}}$]$x_2 = \frac{1}{2}(2 + 0.5) = 1.25 = \frac{5}{4}$ [cite: 516]
- [cite$_{\text{start}}$]$x_3 = \frac{1}{2}(\frac{5}{4} + \frac{4}{5}) = \frac{41}{40}$ [cite: 518]

3.2 2. Proof of Convergence

To use MCT, we check boundedness and monotonicity.

3.2.1 Boundedness (Below)

[cite$_{\text{start}}$]We claim $x_n \geq 1$ for all n[cite: 528]. **Proof:** Using the arithmetic-geometric mean inequality (or completing the square):

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{1}{x_n}\right) \geq \sqrt{x_n \cdot \frac{1}{x_n}} = 1$$

[cite$_{\text{start}}$]Alternatively: $(\sqrt{x_n} - \frac{1}{\sqrt{x_n}})^2 \geq 0 \implies x_n + \frac{1}{x_n} \geq 2 \implies x_{n+1} \geq 1$[cite: 537].

3.2.2 Monotonicity (Decreasing)

We check if $x_{n+1} \leq x_n$:

$$\frac{1}{2}\left(x_n + \frac{1}{x_n}\right) \leq x_n$$

$$x_n + \frac{1}{x_n} \leq 2x_n$$

$$\frac{1}{x_n} \leq x_n \implies 1 \leq x_n^2$$

Since we proved $x_n \geq 1$, this inequality holds. [cite$_{\text{start}}$]Thus, (x_n) is decreasing[cite: 519, 521, 524].

3.3 3. Calculating the Limit

Since the sequence is decreasing and bounded below, it converges to some limit x.

$$\lim_{n\to\infty} x_n = x$$

Applying the limit to the recurrence relation:

$$x = \frac{1}{2}\left(x + \frac{1}{x}\right)$$

[cite$_{\text{start}}$][cite: 539, 541]

$$2x = x + \frac{1}{x} \implies x = \frac{1}{x} \implies x^2 = 1$$

[cite$_{\text{start}}$]Since $x_n > 0$, the limit is $x = 1$[cite: 541].